INCH-POUND

MIL-R-5757/13G
11 April 1990
SUPERSEDING
MIL-R-5757/13F
29 December 1971

MILITARY SPECIFICATION SHEET

RELAYS, ELECTRICAL, HERMETICALLY SEALED, DPDT, LOW LEVEL TO 2 AMPERES, (SENSITIVE, 40 MILLIWATT)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-R-5757.

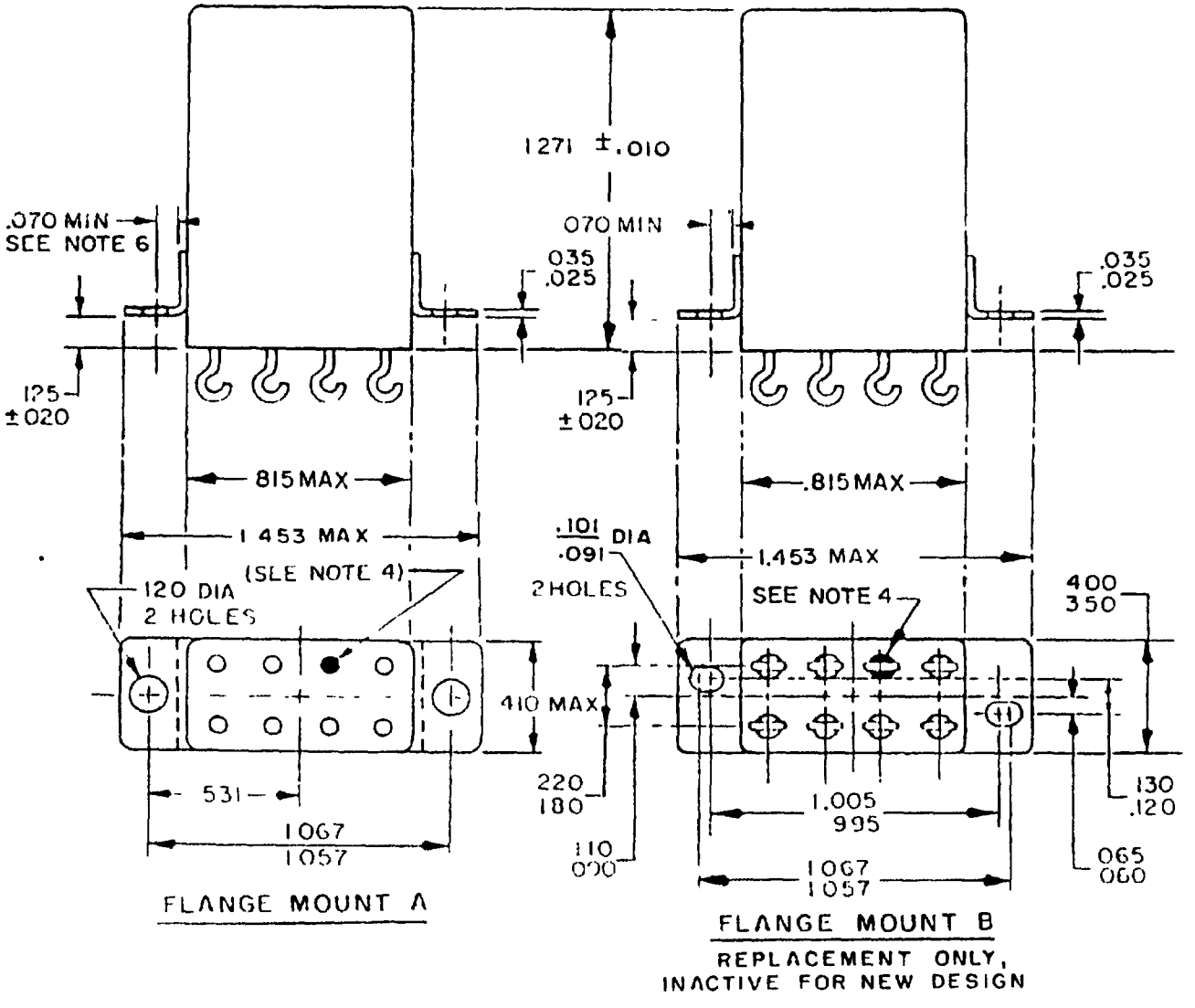


FIGURE 1. Dimensions and configuration.

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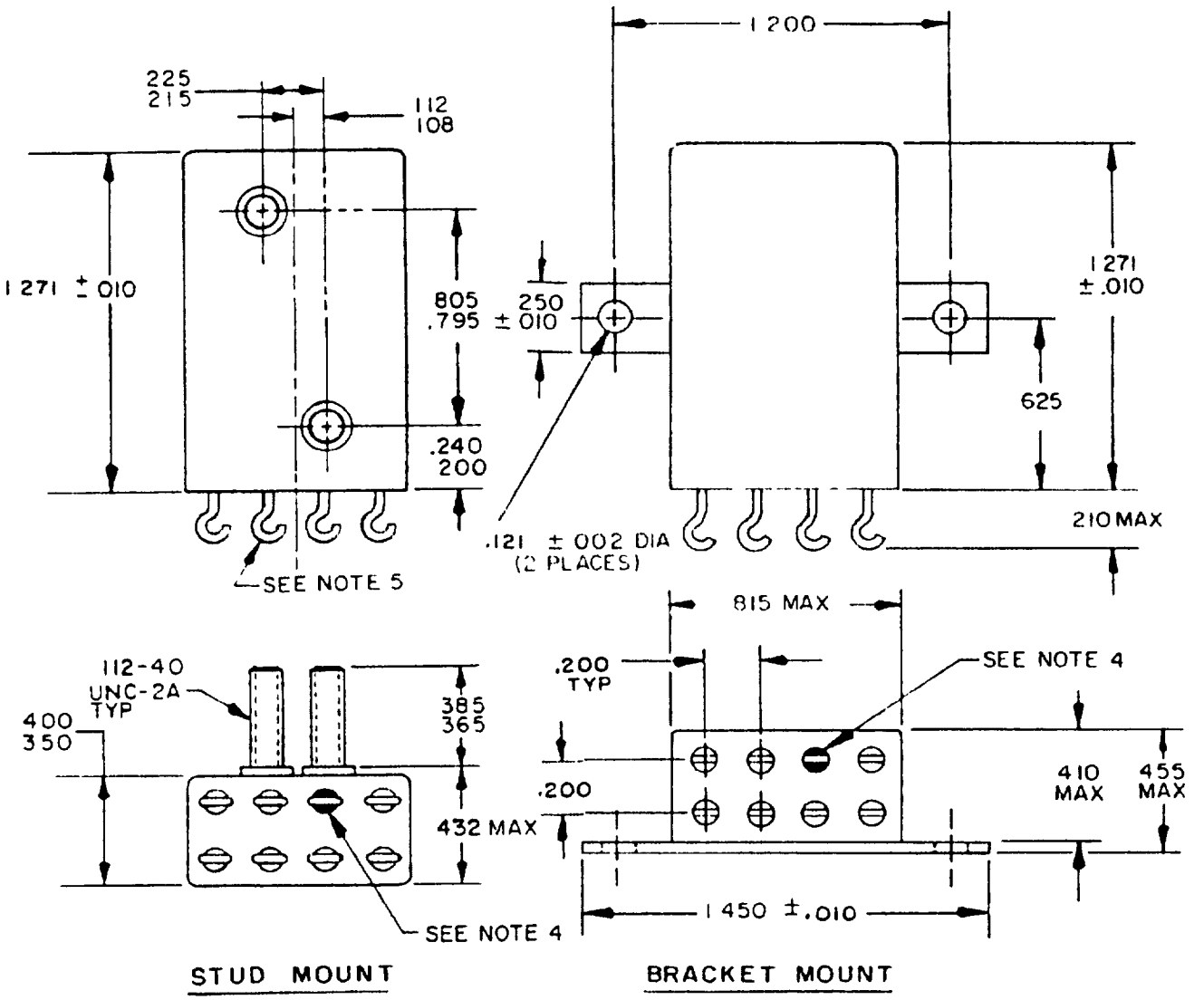
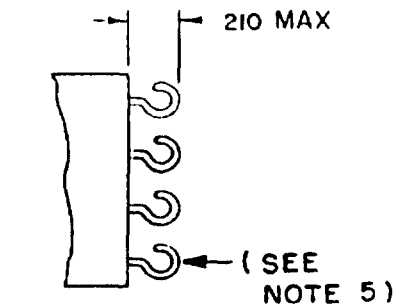
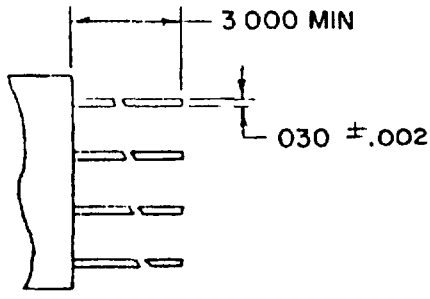


FIGURE 1 Dimensions and configuration - Continued

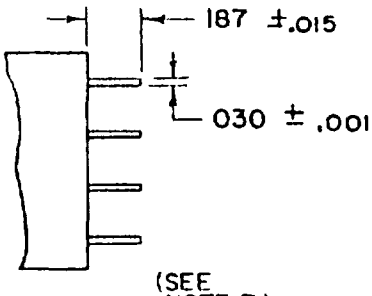
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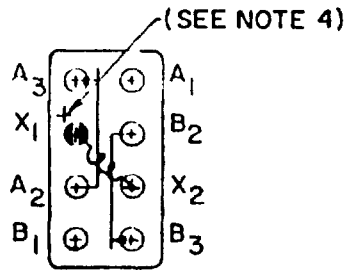
SOLDER LUG



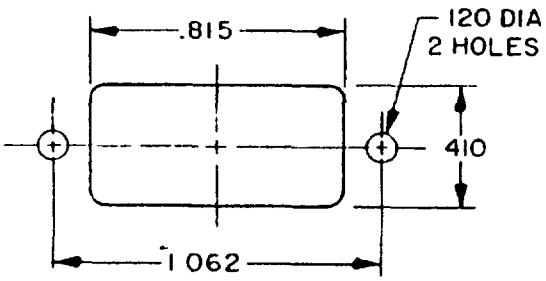
WIRE LEAD



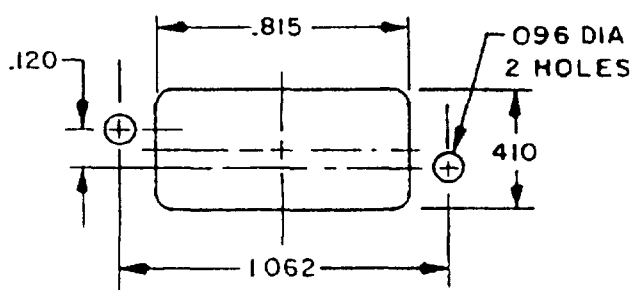
PIN



SCHEMATIC DIAGRAM UNENERGIZED POSITION BOTTOM VIEW



FLANGE MOUNT A
CHASSIS CUTOUT



RECOMMENDED FLANGE MOUNT B
CHASSIS CUTOUT

FIGURE 1 Dimensions and configuration - Continued

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Inches	mm	Inches	mm	Inches	mm	Inches	mm
.001	0.03	.101	2.57	.225	5.72	.815	20.70
.002	0.05	.108	2.74	.240	6.10	.995	25.27
.010	0.25	.110	2.79	.250	6.35	1.005	25.53
.015	0.38	.112	2.84	.350	8.89	1.057	26.85
.020	0.51	.120	3.05	.365	9.27	1.062	26.97
.025	0 64	.121	3.07	.385	9.78	1.067	27.10
.030	0.76	.125	3.18	.400	10.16	1.200	30.48
.035	0.89	.130	3.30	.410	10.41	1.271	32.28
.060	1.52	.180	4.57	.432	10.97	1.450	36.83
.065	1.65	.187	4.75	.455	11.56	1.453	36.91
.070	1.78	.200	5.08	.531	13.49	3.000	76.20
.090	2.29	.210	5.33	.625	15.88		
.091	2.31	.215	5.46	.795	20.19		
.096	2.36	.220	5.59	.805	20.45		

NOTES:
1. Dimensions are in inches.
2. Unless otherwise specified, tolerance is ±.005 (0.13 mm).
3. Metric equivalents are given for general information only.
4. The X_1 terminal shall be identified with a contrasting bead. In the case of relays with polarized coils, the contrasting bead shall indicate the positive terminal and a +(plus) sign shall be placed on the schematic diagram to indicate this terminal.
5. Shape of solder terminals optional.
6. Mounting screw head clearance for .995 (25.27 mm) -1.005 (25.53 mm) dimension based on use of No. 2 filister head screw.
7. Relays with pin terminals shall not be used for plug in socket applications.

FIGURE 1. Dimensions and configuration - Continued.

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REQUIREMENTS

CONTACT DATA

Configuration DPDT.

Arrangement: 2 form C

Load ratings. 1/

High level (relay case grounded):

Resistive: 2 amperes at 28 V dc.

0.3 ampere at 115 V ac, 60 to 400 Hz 2/

Inductive: 0.75 ampere at 28 V dc, at an inductive load of 200 millihenries, minimum

Lamp: 0.1 ampere at 28 V dc.

Low level: 10 to 50 µA at 10 to 50 mV dc or peak ac.

Intermediate current: Applicable.

Contact resistance or voltage drop

Rated life

Before 0.05 ohm

During: 10 percent of open circuit voltage maximum.

After: 0.1 ohm.

Intermediate current.

Before 0.05 ohm

During 3 ohms

After: 3 ohms.

Low level.

Before: 0.05 ohm.

During 100 ohms

After 0.15 ohm

Contact bounce 2 milliseconds (ms) maximum, load shall be 10 mA at 6 V dc maximum.

1/ Each relay possesses high and low level capabilities. However, relays previously tested or used above 10 mA resistive at 6 V dc maximum, or peak ac open circuit, are not recommended for subsequent use in low level applications

2/ For qualifications, testing of 400 hertz (Hz) loads not required

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Overload test

Resistive Two times rated current

Inductive Two times rated current.

COIL DATA.

Duty rating. Continuous.

Nominal current. See table I.

Pick-up current: See table I.

Drop-out current. See table I

Coil release See table I.

Operate time 15 ms maximum over temperature range.

Release time. 10 ms maximum over temperature range.

ELECTRICAL DATA

Insulation resistance 1,000 megohms minimum except the resistance between coil and case at high temperature shall be 500 megohms or greater.

Dielectric withstanding voltage

	Sea level V rms (60 Hz)	Altitude V rms (60 Hz)
Between case, frame, or enclosure and all contacts:	1,000	350 All terminals to case
Between case, frame, or enclosure and coil	500	
Between all contacts and coil.	1,000	
Between open contacts in the energized and unenergized positions	500	
Between contact poles.	1,000	

ENVIRONMENTAL DATA

Temperature range: Symbol B (-65°C to +125°C).

Vibration Symbol 3 (15 G).

Acceleration 30 G.

Shock. Symbol 3 (100 G).

Resistance to soldering heat. Applicable to pin type (printed circuit).

Internal moisture. Applicable.

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PHYSICAL:

Enclosure design: Symbol 4 (hermetically sealed).

Terminal strength: 3 ±0.3 pound pull.

Terminal twist test: Applicable to wire leads.

Sealed by welding: Applicable.

Dimensions and configuration: See figure 1.

Termination: See figure 1 and table I.

Weight 1.1 ounces maximum.

Part or Identifying Number (PIN): See table I.

LIFE TEST REQUIREMENTS:

High level: 100,000 cycles each at rated load.

Two relays per contact rating shall be tested, with rated loads on all contacts, except intermediate current.

For qualification inspection, life lamp load test not required.

Low level: Applicable (100,000 cycles).

Intermediate current. Level II (50,000 cycles).

QUALITY ASSURANCE: (See table II).

Group A.

(1) Subgroup 2: 100 percent.

(2) Dielectric withstanding voltage:

(a) Tests to be conducted at sea level rating only.

(b) Duration of application: 5-10 seconds at a 10 percent increase in the dielectric strength voltage.

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TABLE I. PIN and characteristics.

PIN M5757/13-	Mount	Terminal	DC coil resistance (±10%) +25°C ±5°C	Nominal coil current	Pickup current (max)	Dropout current	
						Max	Min
			(ohms)	(mA)	(mA)	(mA)	(mA)
083	Flange A	Lug (solder)	20.0	90.0	45.0	22.5	4.5
084	Flange A	Lug (solder)	100.0	40.0	20.0	10.0	2.0
085	Flange A	Lug (solder)	500.0	18.0	9.0	4.5	0.9
086	Flange A	Lug (solder)	1,000	13.0	6.5	3.1	0.65
087 1/	Flange A	Lug (solder)	1,500 1/	11.0	5.2	2.6	0.52
088 1/	Flange A	Lug (solder)	2,000 1/	9.0	4.5	2.2	0.50
089	Flange A	Lug (solder)	2,500	8.0	4.0	2.0	0.40
090	Flange A	Lug (solder)	5,000	6.0	2.8	1.4	0.30
091 1/	Flange A	Lug (solder)	8,000 1/	5.0	2.3	1.1	0.23
092	Flange A	Lug (solder)	10,000	4.0	2.0	1.0	0.20
093	Flange A	Pin (PW)	20.0	90.0	45.0	22.5	4.5
094	Flange A	Pin (PW)	100.0	40.0	20.0	10.0	2.0
095	Flange A	Pin (PW)	200.0	28.4	14.2	10.0	1.4
096	Flange A	Pin (PW)	500.0	18.0	9.0	4.5	0.9
097	Flange A	Pin (PW)	1,000	13.0	6.5	3.1	0.65
098 1/	Flange A	Pin (PW)	1,500 1/	11.0	5.2	2.6	0.52
099 1/	Flange A	Pin (PW)	2,000 1/	9.0	4.5	2.2	0.50
100	Flange A	Pin (PW)	2,500	8.0	4.0	2.0	0.40
101	Flange A	Pin (PW)	5,000	6.0	2.8	1.4	0.30
102 1/	Flange A	Pin (PW)	8,000 1/	5.0	2.3	1.1	0.23
103	Flange A	Pin (PW)	10,000	4.0	2.0	1.0	0.20
104 1/	Flange B 1/	Lug (solder)	20.0	90.0	45.0	22.5	4.5
105 1/	Flange B 1/	Lug (solder)	100.0	40.0	20.0	10.0	2.0
106 1/	Flange B 1/	Lug (solder)	500.0	18.0	9.0	4.5	0.9
107 1/	Flange B 1/	Lug (solder)	1,000	13.0	6.5	3.1	0.65
108 1/	Flange B 1/	Lug (solder)	1,500 1/	11.0	5.2	2.6	0.52

See footnote at end of table.

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TABLE I. PIN and characteristics - Continued.

PIN M5757/13-	Mount	Terminal	DC coil resistance (±10%) +25°C ±5°C	Nominal coil current	Pickup current (max)	Dropout current	
						Max	Min
			(ohms)	(mA)	(mA)	(mA)	(mA)
109 1/	Flange B 1/	Lug (solder)	2,000 1/	9.0	4.5	2.2	0.50
110 1/	Flange B 1/	Lug (solder)	2,500	8.0	4.0	2.0	0.40
111 1/	Flange B 1/	Lug (solder)	5,000	6.0	2.8	1.4	0.30
112 1/	Flange B 1/	Lug (solder)	8,000 1/	5.0	2.3	1.1	0.23
113 1/	Flange B 1/	Lug (solder)	10,000	4.0	2.0	1.0	0.20
114	Stud	Lug (solder)	20.0	90.0	45.0	22.5	4.5
115	Stud	Lug (solder)	100.0	40.0	20.0	10.0	2.0
116	Stud	Lug (solder)	500.0	18.0	9.0	4.5	0.9
117	Stud	Lug (solder)	1,000	13.0	6.5	3.1	0.65
118 1/	Stud	Lug (solder)	1,500 1/	11.0	5.2	2.6	0.52
119 1/	Stud	Lug (solder)	2,000 1/	9.0	4.5	2.2	0.50
120	Stud	Lug (solder)	2,500	8.0	4.0	2.0	0.40
121	Stud	Lug (solder)	5,000	6.0	2.8	1.4	0.30
122 1/	Stud	Lug (solder)	8,000 1/	5.0	2.3	1.1	0.23
123	Stud	Lug (solder)	10,000	4.0	2.0	1.0	0.20
124	Stud	Wire lead	20.0	90.0	45 0	22.5	4.5
125	Stud	Wire lead	100.0	40.0	20.0	10.0	2.0
126	Stud	Wire lead	500.0	18.0	9.0	4.5	0.9
127	Stud	Wire lead	1,000	13.0	6.5	3.1	0.65
128 1/	Stud	Wire lead	1,500 1/	11.0	5.2	2.6	0.52
129 1/	Stud	Wire lead	2,000 1/	9.0	4.5	2.2	0.50
130	Stud	Wire lead	2,500	8.0	4.0	2.0	0.40
131	Stud	Wire lead	5,000	6.0	2.8	1.4	0.30
132 1/	Stud	Wire lead	8,000 1/	5.0	2.3	1.1	0.23
133	Stud	Wire lead	10,000	4.0	2.0	1.0	0.20

See footnote at end of table.

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TABLE I. PIN and characteristics - Continued.

PIN M5757/13-	Mount	Terminal	DC coil resistance (±10%) +25°C ±5°C	Nominal coil current	Pickup current (max)	Dropout current	
						Max	Min
			(ohms)	(mA)	(mA)	(mA)	(mA)
134	Bracket	Lug solder	20.0	90.0	45.0	22.5	4.5
135	Bracket	Lug solder	100.0	40.0	20.0	10.0	2.0
136	Bracket	Lug solder	500.0	18.0	9.0	4.5	0.9
137	Bracket	Lug solder	1,000	13.0	6.5	3.1	0.65
138 1/	Bracket	Lug solder	1,500 1/	11.0	5.2	2.6	0.52
139 1/	Bracket	Lug solder	2,000 1/	9.0	4.5	2.2	0.50
140	Bracket	Lug solder	2,500	8.0	4.0	2.0	0.40
141	Bracket	Lug solder	5,000	6.0	2.8	1.4	0.30
142 1/	Bracket	Lug solder	8,000 1/	5.0	2.3	1.1	0.23
143	Bracket	Lug solder	10,000	4.0	?.0	1.0	0 20

1/ Inactive for new design

TABLE II. Qualification inspection and sample size.

Single submission	Group submission	
20 units plus 1 open unit Table VIII Qualification inspection as applicable.	M5757/13-092	20 units plus 1 open unit Qualification inspection as applicable.
	M5757/13-113 M5757/13-133 M5757/13-143	2 units each PIN Qualification inspection Group I and shock, vibration, acceleration, terminal strength, and seal.
	M5757/13-103	4 units Qualification inspection and shock, vibration, acceleration, terminal strength, resistance to soldering heat, and seal.

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SUPERSESSION DATA·

Supersession data See table III.

TABLE III. Supersession data.

Superseded PIN M5757/13-	New PIN M5757/13-	Superseded PIN M5757/13-	New PIN M5757/13-
011	104 1/	054	097
012	105 1/	055	098 1/
013	106 1/	056	099 1/
014	107 1/	057	100
015	108 1/	058	101
016	109 1/	059	102 1/
017	110 1/	060	103
018	111 1/	061	124
019	112 1/	062	125
020	113 1/	063	126
031	114	064	127
032	115	065	128 1/
033	116	066	129 1/
034	117	067	130
035	118 1/	068	131
036	119 1/	069	132 1/
037	120	070	133
038	121	072	134
039	122 1/	073	135
040	123	074	136
041	083	075	137
042	084	076	138 1/
043	085	077	139 1/
044	086	078	140
045	087 1/	079	141
046	088 1/	080	142 1/
047	089	081	143
048	090	082	095
049	091 1/		
050	092		
051	093		
052	094		
053	096		

1/ Inactive for new design.

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Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

CONCLUDING MATERIAL

Custodians:
Army - ER
Navy - EC
Air Force - 85

Review activities:
Army - AR, AT, MI
Navy - AS, OS, SH
Air Force - 11, 17
DLA - ES

User activities
Army - AV, ME
Navy - MC
Air Force - 19

Preparing activity:
Navy - EC

Agent
DLA - ES

(Project 5945-0801)

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